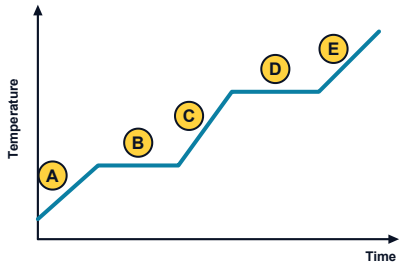
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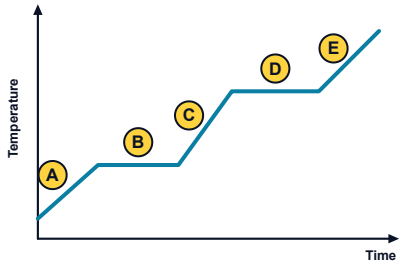
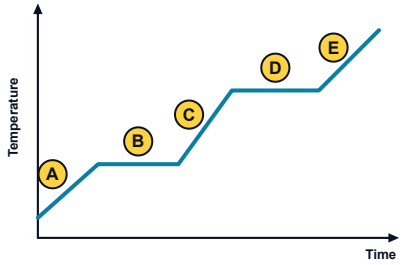
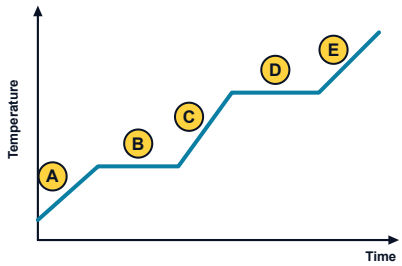
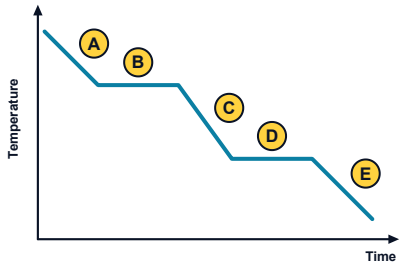
Changes of state and specific latent heat

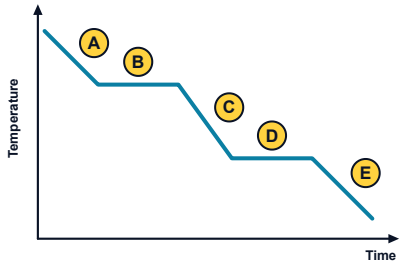
Particle model of matter
Paper 1, Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the standard unit of specific latent heat.**	Joule per kilogram (J/kg)	☐
2 ★	**Define latent heat.**	The energy needed to change the state of a substance with no change in temperature.	☐
3 ★	**Define specific latent heat.**	The energy needed to change the state of 1 kg of a substance with no change in temperature.	☐
4	**Define specific latent heat of fusion.**	The energy needed to change 1 kg of a substance from solid to liquid with no change in temperature.	☐
5	**Define specific latent heat of vaporisation.**	The energy needed to change 1 kg of a substance from liquid to vapour (gas) with no change in temperature.	☐
6	**Which letter on the heating curve shows the substance melting?** 	B	☐

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	QUESTION	ANSWER	✓
7	**Which letter on the heating curve shows the substance boiling?** 	D	☐
8	**Which letter on the heating curve shows a solid getting hotter?** 	A	☐
9	**Which letter on the heating curve shows a liquid getting hotter?** 	C	☐
10	**Which letter on the cooling curve shows the substance freezing?** 	D	☐

	QUESTION	ANSWER	✓
11	**Which letter on the cooling curve shows a liquid cooling?** 	C	☐
12	**State the equation linking energy for a change of state, mass and specific latent heat.**	energy for a change of state = mass × specific latent heat (E = m L)	☐
13 ★	**Explain why the temperature stays constant while a substance melts.**	The energy supplied increases the potential energy of the particles, not their kinetic energy.	☐
14	**State the difference between specific heat capacity and specific latent heat.**	Specific heat capacity: energy to change the temperature of 1 kg by 1 °C. Specific latent heat: energy to change the state of 1 kg.	☐